

Mustafa Mert Çelikok

Assistant Professor, Multi-Agent Reinforcement Learning

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🔗 Google Scholar

Summary

Assistant Professor at the University of Southern Denmark, working on multi-agent reinforcement learning, cooperative AI, and Bayesian/model-based approaches to human-AI interaction. PhD from Aalto University (2023), postdoc at TU Delft (2023-2025).

Education

PhD	Aalto University , Computer Science	Finland
	- Thesis: Model-based Multi-agent Reinforcement Learning for AI Assistants - Advisor: Professor Samuel Kaski - Opponent: Professor Matthew A. Taylor (University of Alberta) - Nominated for the Dean's Award	2019 – 2023
MSc	Aalto University , Machine Learning, Data Science and Artificial Intelligence (MACADAMIA)	Finland
	Completed with honours (highest distinction)	2016 – 2018
BEng	Selçuk University , Computer Engineering	Turkey
	First-class honours (highest distinction)	2011 – 2015

Academic Appointments

University of Southern Denmark, Department of Mathematics and Computer Science, Data Science Section , Tenure-track Assistant Professor	Denmark
	2025 – present
Delft University of Technology , Postdoctoral Researcher	The Netherlands
Project on investigating advanced multi-agent reinforcement learning techniques for collaborative AI systems, funded for 3 years as part of the Dutch Hybrid Intelligence Centre.	2023 – 2025
Dutch National Research Institute for Mathematics and Computer Science (CWI) , PhD Research Intern	The Netherlands
Highly selective doctoral internship programme with stipend. Worked with Professor Wouter Koolen on the sample-complexity of Monte Carlo Tree Search.	Jan 2023 – Apr 2023

Teaching Experience

Spring School on Reinforcement Learning and Control, CWI , Teaching Staff and Co-organizer	2025 – 2025
Ran tutorials on reinforcement learning for an interdisciplinary group of students, plus organizational duties.	
ELLIS Summer School on Collaborative and Generative AI , Invited Tutorial Speaker	2024 – 2024
SIKS Course on RL for Adaptive Hybrid Intelligence , Invited Lecturer	2023 – 2023
Invited lecture on reinforcement learning for ad-hoc cooperation, part of a PhD course in the Netherlands Research School for Information and Knowledge Systems.	

CS-E4070 - Probabilistic Modelling for Cognition and Interaction, Aalto University , Co-organizer and Instructor	2018 – 2018
Designed and implemented a Master's and PhD level special topic course together with Dr. Tomi Peltola.	

Supervision

Mia Choi — MSc in Cognitive Robotics, TU Delft	2024 – 2024
"Multi-Agent Reinforcement Learning for Portfolio Management", in collaboration with Robeco Investment.	
Aranya Sinha — MSc in Computer Science, TU Delft	2024 – 2024
"Risk-sensitive Reinforcement Learning for Portfolio Allocation", in collaboration with Robeco Investment.	
Alex Hämäläinen — MSc in Computer Science, Aalto University	2021 – 2021
"Differentiable User Models"	
Ari-Pekka Härkönen — BSc in Computer Science, Aalto University	2019 – 2019
"Computational Models of Theory of Mind"	

Grants and Awards

ELLIS PhD Student	2019
Nominated and accepted to the pan-European PhD programme of the European Lab for Learning and Intelligent Systems, as part of its first cohort.	
KAUTE Personal Research Grant	2019
Personal doctoral grant from the Finnish Science Foundation for Technology and Economics, received four times in a row (2019-2023), totalling 96,000 euros.	
Aalto University Competitive Co-department Funding Recipient	2020
PhD project granted co-department funding as part of an internal process.	

Academic Leadership and Service

Danish Pioneer Centre for AI Programme on Learning and Optimal Control in Dynamical Systems , Co-director	2026 – 2028
ICLR Workshop: Multi-Agent Learning and Its Opportunities in the Era of Generative AI , Co-organizer	2026 – 2026
AAAI Bridge Workshop: Collaborative AI and Modelling of Humans , Co-organizer	2024 – 2025
Co-organized both editions of this workshop at the AAAI Conference on Artificial Intelligence; the 2024 edition reached 300 registrations.	
Hybrid Intelligence Centre Reinforcement Learning SIG , Co-leader	2023 – 2025
Program Committee Member: NeurIPS, ICML, ICLR, AAAI, AAMAS, STOC, BNAIC, JMLR, JAIR , Reviewer	

Invited Workshops and Seminars

Lorentz Centre Workshop: Research Environments for Human-Machine Teaming , Invited Attendee	Leiden, The Netherlands 2024 – 2024
Dagstuhl Workshop: Model Learning for Improved Trustworthiness in Autonomous Systems , Invited Attendee	Schloss Dagstuhl, Germany 2023 – 2023

Selected Publications

Distributional Active Inference

2026

Alperen Akgül, Gecia Baykal, Manuel Hausmann, Mustafa Mert Çelikok, Melih Kandemir
openreview.net/forum?id=m3O4qYQ5hu (Forty-third International Conference on Machine Learning (ICML))

On the Complexity of Learning to Cooperate with Populations of Socially Rational Agents

2025

Sreejato Bandyopadhyay, Mustafa Mert Çelikok, Robert Loftin
(Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS))

Value Improved Actor Critic Algorithms

2025

Yaniv Oren, Moritz A. Zanger, Pascal R. Van der Vaart, Mustafa Mert Çelikok, Matthijs T. J. Spaan, Wendelin Böhmer
(Advances in Neural Information Processing Systems (NeurIPS))

Uncoupled Learning of Differential Stackelberg Equilibria with Commitments

2024

Robert Loftin, Mustafa Mert Çelikok, Herke Hoof, Samuel Kaski, Frans A. Oliehoek
(Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS))

Best-Response Bayesian Reinforcement Learning with Bayes-adaptive POMDPs for Centaurs

2022

Mustafa Mert Çelikok, Frans A. Oliehoek, Samuel Kaski
(Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems (AAMAS))

Model-based Multi-agent Reinforcement Learning for AI Assistants

2023

Mustafa Mert Çelikok
aaltodoc.aalto.fi/handle/123456789/120725 (PhD thesis, Aalto University)